Miniproject 3, cool-down
Epidemics Simulation

October 2026

Introduction

There are a number of mathematical models describing how epidemics spread in a population. Simulation of disease spread can for example give information on how fast the disease might spread and how different measures, such as vaccination, might affect the spread. It might also give the health care system crucial information that affect the number of intensive care units and the planning in general.

Background

The mathematical model we will use here is the SEIR model, schematically described in Figure 1. The population is divided into compartments:

- S – **S**usceptible, individuals who have not yet caught the disease and can become infected.
- E – **E**xposed, individuals who have been infected by the pathogen but are not yet infectious or showing symptoms (the incubation period).
- I – **I**nfectious, individuals who are actively infected (and usually ill) and can transmit the disease to susceptible people
- R – **R**ecovered, individuals who have recovered from the illness, gained immunity or have passed away

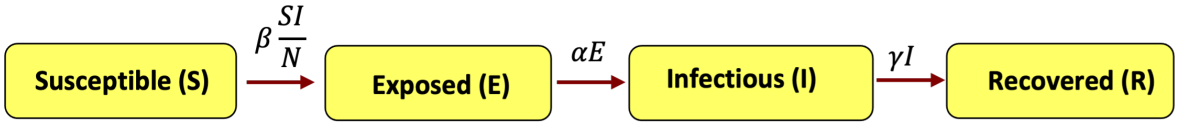


Figure 1: The SEIR model in epidemiology

The SEIR-model can be written as an ordinary differential equation (ODE):

$$
\begin{aligned}
S'(t) &= -\tfrac{\beta I}{N} S \\
E'(t) &= \tfrac{\beta I}{N} S - \alpha E \\
I'(t) &= \alpha E - \gamma I \\
R'(t) &= \gamma I
\end{aligned}
$$

This is the deterministic version of the model. The parameters in the model are

- N – the population size. It's always the case that $N = S + E + I + R$, i.e. individuals always belong to one of the states.
- β – the transmission rate, the average number of adequate contacts per person per unit of time, multiplied by the probability of disease transmission upon contact.
- γ – the fraction of infected individuals who recover (with immunity) or are removed from the infected pool per unit of time.
- α – the rate at which exposed individuals transition into the infectious state, $\frac{1}{\alpha}$ is the incubation period

The so called *basic reproductive number*, often denoted R_0 is in this model $R_0 = \frac{\beta}{\gamma}\frac{S(0)}{N}$.

You can read more about the model on Wikipedia [1]

To Do

1. Simulate the epidemic using the deterministic (ODE) model. Here, use the SciPy method `scipy.integrate.solve_ivp` (you've probably used it in a previous course). Use $N = 100$ (we can think in percent here), and let $\beta = 0.6$, $\alpha = 1/5$ (5 days incubation period), $\gamma = 1/10$ (infectious period of 10 days). These parameter values are roughly the ones that were derived from empirical averages of the SARS-CoV-2 virus. Assume that 2 individuals (or 2%) are infectious at time zero, i.e. $I(0) = 2$, zero exposed and recovered (and note that $N = S + E + I + R$). Run the simulation for 60 days, and plot the result. You can use this solution as a reference when working with the stochastic model.

   Include calculation of the R_0-number in your code, and print it when you run the code. Reduce β until you reach $R_0 = 1$. What happens with the number of infectious when $R_0 = 1$ (or < 1)?

[1] https://en.wikipedia.org/wiki/Compartmental_models_(epidemiology).

2. From the ODE-model, formulate the corresponding stochastic model. Use the Python-function `SSA.py` (found in the file `Gillespe.py`) on the Studium page, and run the same problem as in previous task (when $\beta = 0.6$).
   Compare the results with the ODE-solutions. They should resemble each other, but not be the same. If you compare with the result in task 1, what are the fundamental differences between the deterministic vs. stochastic simulations?
   (Note, some simulations might die out. This is not wrong, but a consequence of randomness).

3. Here, we will only focus on the number of infectious (variable I). In the same figure, plot variable I from the ODE-solution and a few SSA-simulations (you can for example try $15 - 30$ SSA-simulations). In relation to the law of large numbers, does it seem to be correct?

4. Assume that the health care system suspect big problems if the number of infectious reach 40 % or higher. Therefore, they would like to know the probability for that scenario to happen. How would you be able to do simulations that answer that question? Write a code and do this simulation.
   (The probability won't be exact, but you should be able to give a rough estimate).